

Object Concepts Emerge from Motion

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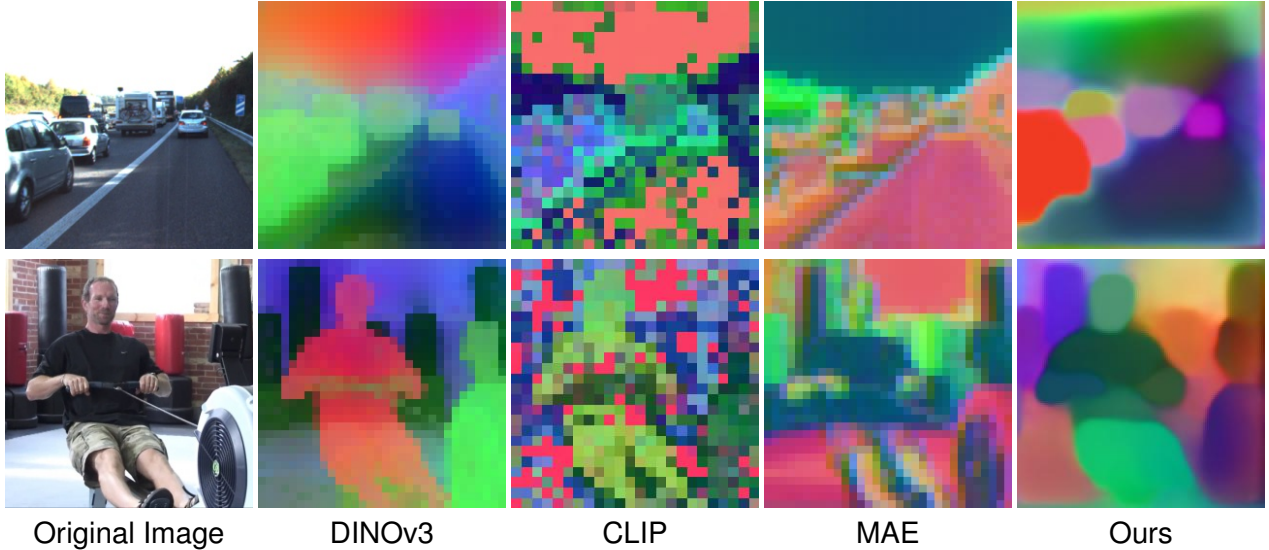


Fig. 1. Motion teaches a single-image encoder which pixels belong to each object, internalizing dynamic grouping into static features. Compared with DINOv3, CLIP, and MAE, ours preserves within-object coherence while separating distinct instances. This instance-aware structure supports geometry and instance-sensitive scene understanding.

Abstract—Object concepts play a foundational role in human visual cognition, enabling perception, memory, and interaction in the physical world. Motion, in turn, provides complementary cues about object unity, extent, and instance separation that are not directly available from static appearance alone. Inspired by findings in developmental neuroscience, where infants acquire object understanding through the observation of motion, we propose a biologically inspired framework for learning object-centric visual representations from raw videos. Our key insight is that motion boundaries provide a strong signal for object-level grouping and can be used to derive pseudo-instance supervision. Concretely, we generate motion-derived instance masks using off-the-shelf optical flow and clustering, and use them to train a single-image encoder through pixel-level pairwise metric learning. Our framework requires no human annotations for representation pretraining or camera calibration, making it scalable to large-scale unstructured video data. We scale motion-derived pretraining to 195M pseudo-labeled frames and introduce Motion-Verified Self-Training, which combines model proposals with motion cues to expand supervision to 421M pseudo-labeled frames from the same videos. We further scale the encoder to Swin-H and distill it into compact variants. We evaluate the learned representations on four downstream tasks spanning low-level and high-level vision: monocular depth estimation, 3D object detection, 3D occupancy prediction, and end-to-end planning. Our models achieve competitive or superior transfer relative to supervised and self-supervised pretraining baselines, with particularly strong results on geometry- and instance-sensitive tasks. These results suggest that motion-induced object representations provide a distinct object-centric inductive bias for visual pretraining, capturing a crucial but overlooked level of abstraction: the visual instance.

Index Terms—Object-centric learning, self-supervised learning, optical flow, video pretraining, visual foundation models.

1 INTRODUCTION

PHYSICAL AI aims to develop intelligent agents capable of perceiving and interacting with the physical world. A fundamental cognitive capacity required by such agents is the ability to recognize and understand the concept of an *object*, a core unit of perception and reasoning. In the human visual system, the importance of object concepts is well established in cognitive science: object-level abstraction supports not only recognition and classification, but also spatial perception, memory formation, and reasoning about

interactions between objects and their environment [1], [2]. Understanding how object concepts are formed in biological systems therefore provides valuable guidance for building more robust and generalizable artificial visual systems.

However, what makes an object look like an object? This is a non-trivial question, as objects vary drastically in appearance, shape, and motion patterns. Studies in developmental psychology have shown that the ability to perceive object unity is not innate but learned during infancy: infants begin to exhibit an understanding of object cohesion at around two months of age, with robust performance by four

months [1]. Crucially, motion cues, particularly common or coherent motion, serve as the dominant signal from which infants infer object boundaries and unity [2], [3]. As the visual system matures, this dynamic understanding is gradually internalized into the ventral visual stream [4], enabling object recognition from static visual input alone. This developmental trajectory suggests a natural computational recipe: use motion to teach a model what objects are, and let the resulting knowledge be exercised on single static images.

This recipe addresses a gap in current visual pretraining. Self-supervised and weakly supervised foundation models such as DINO [5], [6], MAE [7], and CLIP [8] learn broadly transferable representations from massive data, and their features have become the default starting point for a wide range of vision tasks. Yet when we inspect what these models capture (Fig. 1), their features predominantly encode *semantic class* or texture-level structure: pixels of the same category are mapped close together, whereas the boundary between two adjacent instances of the same category is largely invisible in feature space. Instance-level organization, in which each object is represented as a distinct and coherent visual entity, is often underrepresented. We argue that this level of abstraction is fundamental for physical scene understanding, where an agent must reason about individual objects, their extents, and their geometry.

Motivated by these observations, we propose a biologically inspired framework that learns object-centric representations directly from raw videos. The key geometric fact underlying our method is that motion boundaries often align with object boundaries: an independently moving object separates from its surroundings in the optical flow field, and even in rigid scenes, camera ego-motion induces flow discontinuities exactly at depth discontinuities, which frequently coincide with physical object boundaries. We therefore run an off-the-shelf optical flow estimator on unlabeled videos, group pixels with coherent flow into motion-derived pseudo instance masks by simple clustering, and train a dense image encoder with a pixel-level pairwise metric-learning objective: pixels within the same pseudo instance are pulled together in feature space, while pixels from different instances are pushed apart. The representation-pretraining pipeline requires no human annotations on the target video corpus, camera calibration, multi-view input, or LiDAR input, so it scales naturally to large, unstructured video collections. Motion is used only during pretraining; the resulting encoder consumes a single image and serves as an ordinary visual backbone.

A preliminary version of this work [9], trained on 48M pseudo-labeled frames from driving videos, outperformed supervised ImageNet-22K pretraining and strong self-supervised baselines on monocular depth estimation, 3D object detection, and 3D occupancy prediction. This article pushes the paradigm further, organized around a single question: *how far does motion-derived object supervision scale?* We study three axes: the size and diversity of the video corpus, the amount of reliable supervision extracted from each video, and the capacity of the pretrained model. In particular, the conference pipeline favors label precision by retaining only sufficiently complete flow clusters, leaving useful but incomplete motion evidence unexploited when boundaries are broken by noise or occlusion. This motivates

a second training cycle that combines model proposals with motion constraints to increase label coverage.

We extend the framework along all three axes. *Scaling the data.* A unified processing pipeline produces 195M frames with motion-derived pseudo labels from 7,163 hours of heterogeneous driving and self-collected videos. *Scaling the supervision.* Inspired by the data-engine principle of SAM 2 [10], Motion-Verified Self-Training combines proposals from the cycle-1 encoder with motion evidence generated independently of those proposals, expanding usable supervision to 421M pseudo-labeled frames from the same video sources. *Scaling the model.* We train a Swin-H backbone and distill its representation into a family of compact models.

Across the four downstream tasks, the extended framework achieves strong transfer: it outperforms matched baselines on monocular depth estimation, 3D object detection, and 3D occupancy prediction, while remaining competitive on end-to-end planning. On KITTI depth estimation, the current Swin-L improves SILog from 6.558 in the conference version to 5.736, while the Swin-H encoder reaches 5.843; on nuScenes 3D detection, the current Swin-L improves NDS from 55.80 to 55.89, while Swin-H reaches 57.48; the current Swin-L also improves occupancy RayIoU from 38.70 to 40.04; and end-to-end planning shows competitive transfer under NAVSIMv2. The expanded pseudo-label coverage and the improved downstream results together demonstrate the effectiveness of the complete Cycle-2 pipeline. A controlled Swin-T comparison further strengthens this conclusion: with a slightly smaller additional sample-exposure budget, 10 epochs on the Cycle-2 labels outperform extending Cycle-1 pretraining from 50 to 75 epochs across all non-saturated depth metrics. This controlled result demonstrates that switching to the improved Cycle-2 supervision yields better transfer than simply training longer on the original labels.

Our contributions are summarized as follows:

- We present a biologically inspired paradigm for object-centric visual representation learning that converts optical flow into pseudo instance supervision and requires neither human annotations on the target video corpus nor camera calibration, and we demonstrate its effectiveness and scalability on modern architectures.
- We scale motion-derived supervision to 195M pseudo-labeled frames from 7,163 hours of heterogeneous driving and self-collected videos through a unified large-scale processing pipeline.
- We propose Motion-Verified Self-Training, which combines model proposals with proposal-independent motion evidence to expand usable supervision to 421M pseudo-labeled frames from the same video sources and further improve the depth, detection, and occupancy evaluations under a fixed Swin-H backbone; a controlled Swin-T experiment demonstrates more effective transfer than continued pretraining on the original labels under a comparable budget.
- We scale the backbone to Swin-H, distill it into a family of compact models, and extensively evaluate

on four downstream tasks, monocular depth estimation, 3D object detection, 3D occupancy prediction, and end-to-end planning, where our models achieve competitive or superior transfer relative to supervised and self-supervised baselines.

Relative to that conference version, this journal article scales motion-derived pretraining from 48M to 195M frames, introduces Motion-Verified Self-Training to obtain 421M pseudo-labeled frames (Sec. 3.4), scales the backbone to Swin-H with compact-model distillation, and adds end-to-end planning and broader experimental analyses (Sec. 4).

2 RELATED WORK

2.1 Object Discovery

Object discovery aims to identify and localize objects without instance-level annotations for predefined categories. Early methods exploit repeated object occurrence across images [11]–[13] or select object regions from proposal sets through combinatorial optimization [14]–[18]. More recent approaches use features from pretrained image encoders, such as DINO, together with graph partitioning or spectral clustering to localize and segment objects [19]–[22]. Object-centric generative models instead decompose a scene into entities by learning to reconstruct or generate its constituent objects and background [23]–[26].

Motion and multimodal observations provide another route to object discovery. These methods use the motion consistency of 2D or 3D points to separate objects from their surroundings [27], [28]. Our work also extracts object regions from motion, but object discovery is not the final task. We use optical flow and clustering to construct category-agnostic pseudo instance masks at scale, which then supervise a single-image representation. This separates the source of supervision, video motion, from the deployed encoder, which requires only a static image.

2.2 Learning Representations from Motion

Motion can supervise visual representation learning without being the representation target at inference time. Early work uses ego-motion as a supervisory signal, connecting visual features to the movement of the observer [29]. Pathak et al. [30] obtain foreground masks from unsupervised motion segmentation and train a ConvNet to predict them, demonstrating that motion-derived grouping can transfer to recognition. More recently, Divided Attention (DivA) [31] discovers multiple independently moving regions with slot-based flow grouping and uses the discovered proposals to bootstrap a static DINO representation. We share this motion-to-representation principle, but use motion-derived multi-instance masks as dense pairwise supervision, directly organizing pixel features around object unity and instance separation. Our pipeline requires neither ego-motion measurements nor camera calibration, scales to 421M pseudo-labeled frames and modern hierarchical backbones, and uses motion-verified self-training to recover supervision discarded by conservative flow clustering.

Motion is also central to unsupervised video object segmentation, where the objective is to produce object

masks from video. Representative methods discover moving objects through adversarial contextual reasoning [32], appearance-motion decomposition [33], slot-based flow grouping [34], or motion anticipation [35]. In our framework, the masks are intermediate supervision rather than the final output: their utility is evaluated through the downstream transfer of the image representation they induce.

Video self-supervised methods such as VideoMAE [36] and V-JEPA [37] learn spatiotemporal representations by reconstructing or predicting masked video content. They capture motion implicitly through temporal context and typically produce encoders designed for video clips. In contrast, we explicitly convert motion into object-level grouping supervision and train an encoder that operates on a single image at inference time.

2.3 Visual Foundation Models

Visual foundation models learn transferable representations from large-scale data using several forms of pretraining. Contrastive methods enforce invariance across augmented views [38]–[43], while self-distillation methods obtain supervisory targets from a teacher network without requiring negative pairs [5], [6], [44], [45]. Masked-image modeling learns by reconstructing missing image content [7], [46]–[49], and joint embedding-prediction objectives predict representations rather than pixels [37], [50]. Vision-language models such as CLIP additionally exploit image-text supervision to acquire broad category semantics [8]. DINOv3 further scales self-distillation while explicitly improving dense visual features [51].

Large-scale pretraining has also been specialized for structured visual outputs. Promptable segmentation models such as SAM and SAM 2 learn broadly applicable mask prediction from large segmentation corpora [10], [52]. Our method instead uses motion as a category-agnostic grouping signal, encouraging pixels within an object to remain coherent and adjacent instances to remain distinguishable. As suggested by Fig. 1, this inductive bias differs from objectives dominated by category invariance or appearance reconstruction, and is particularly relevant to geometry- and instance-sensitive downstream tasks.

2.4 Self-Training and Pseudo-Label Scaling

Self-training exploits unlabeled data by using a seed model to generate pseudo labels for subsequent training [53], [54]. Confidence filtering, consistency regularization, and teacher-student targets can improve the reliability of this process [55], [56]. Quality control is especially important for dense supervision: boundary errors, merged instances, and missing object parts affect many pixels and can be amplified when model predictions are reused as training targets.

Large segmentation systems address this issue through iterative data engines. SAM uses a model-assisted annotation loop to construct a large mask corpus [52]. SAM 2 extends this strategy to video: progressively stronger models assist human annotators, propose additional masks, and help determine which annotations require correction before entering the training set [10]. The broader lesson is that scaling pseudo labels requires a verification signal rather than accepting model predictions wholesale.

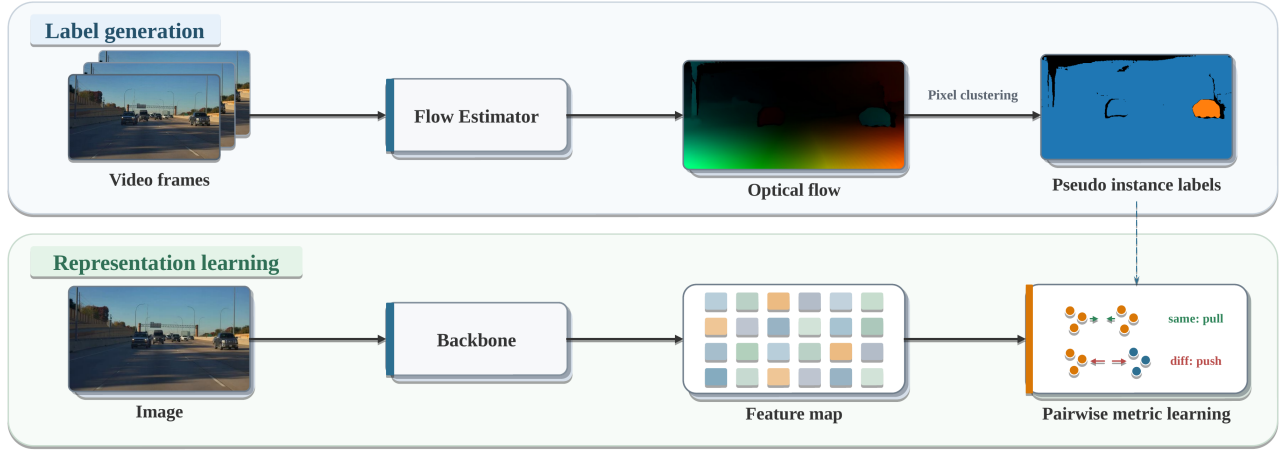


Fig. 2. Cycle-1 motion-derived object supervision. The upper row converts raw video frames into pseudo instance labels: a flow estimator produces optical flow, and pixel clustering over reliable flow regions yields pseudo instance labels. The lower row trains an image encoder from these labels: a single image is encoded into dense features and optimized with pairwise metric learning, pulling pixels from the same pseudo instance together and pushing pixels from different instances apart.

Our Motion-Verified Self-Training adopts this principle without manual verification on the target video corpus. Model proposals increase mask coverage, while proposal-independent motion evidence verifies and refines them. Motion therefore remains an external constraint on the self-training loop, distinguishing our approach from conventional self-training based only on prediction confidence or teacher-student agreement.

3 METHOD

3.1 Overview

Our method learns an object-centric image representation from raw videos without manual instance annotations. The central supervision signal is motion: coherent motion suggests object unity, while motion discontinuities often reveal object boundaries. Motion is used only during pretraining; after training, the encoder takes a single image as input and serves as a standard visual backbone.

The framework has two training cycles. In Cycle 1, shown in Fig. 2, we convert optical flow into conservative motion-derived pseudo instance labels and train dense image features with a pairwise metric-learning objective. We scale this supervision across heterogeneous video sources with a unified processing and filtering pipeline, yielding 195M pseudo-labeled frames. In Cycle 2, shown in Fig. 4, we address the limited coverage of conservative flow clustering through Motion-Verified Self-Training: the cycle-1 encoder proposes candidate masks, while motion cues verify and refine them before they become cycle-2 pseudo labels. The encoder is then fine-tuned on the resulting set of 421M motion-verified pseudo-labeled frames, with its effect evaluated in Sec. 4.

3.2 Motion-Derived Object Supervision

Motion boundaries provide a geometric cue for object boundaries. When an object moves independently, its image motion separates from the surrounding scene. Camera

translation can also turn depth discontinuities into flow discontinuities in rigid scenes, and these discontinuities often align with physical object boundaries. This link lets us convert raw video into object-level supervision without human annotations on the target videos or camera calibration.

We formalize this intuition under a pinhole camera model. Let $p = (u, v)$ denote an image pixel, $D(u, v)$ denote its depth, and K denote the intrinsic matrix. The pixel corresponds to a 3D point $P = D(u, v)K^{-1}[u, v, 1]^\top$. Under a rigid camera motion $(R, t) \in SE(3)$, this point projects to a new pixel $p' = (u', v')$ in the next frame:

$$\begin{bmatrix} u' \\ v' \\ 1 \end{bmatrix} \sim K \left(RD(u, v)K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} + t \right). \quad (1)$$

The optical flow at p is therefore a function of pixel location, depth, camera motion, and camera intrinsics:

$$\mathbf{w}(p) = \begin{bmatrix} u' - u \\ v' - v \end{bmatrix} = \phi(p, D(p); R, t, K). \quad (2)$$

Let p_0 lie on a depth boundary separating two regions Ω^- and Ω^+ . We use $D^\pm$ and $\mathbf{w}^\pm$ to denote the one-sided depth and optical-flow limits, respectively, as p approaches p_0 from within $\Omega^\pm$. Under the same rigid camera motion on both sides, the flow limits and their jump are

$$\mathbf{w}t_{p_0} = \phi(p_0, D^+; R, t, K) - \phi(p_0, D^-; R, t, K). \quad (3)$$

When the camera motion contains a non-degenerate translational component, the projected displacement generally depends on depth; consequently, $D^+ \neq D^-$ can induce $\mathbf{w}^+ \neq \mathbf{w}^-$. Independently moving objects provide another source of flow discontinuities because their motion differs from that of the surrounding scene. Motion boundaries therefore provide a practical, though not one-to-one, proxy for object boundaries, while the optical flow field already encodes the required geometric information.

We convert this geometric cue into pseudo instance masks by grouping pixels whose flow vectors form coherent regions.

Network architecture. The representation network takes a single image as input and consists of a backbone, a Feature Pyramid Network (FPN), and a dense prediction head based on the semantic branch of Panoptic FPN [57]. The FPN fuses multiscale backbone features, and the prediction head produces a 64-channel dense feature map. We ℓ_2 -normalize each sampled pixel feature before applying the pairwise metric-learning objective.

Pixel-level objective. The pseudo instance masks supervise the network through a pixel-level metric-learning objective. For each augmented training crop, we sample 200 pixels from the valid-flow regions defined by the forward-backward consistency check in Sec. 3.3, distributing them approximately evenly across the pseudo labels present in the crop. We form all within-image pairs among these sampled pixels and optimize their feature distances according to the corresponding pseudo labels. Pixels from the same foreground cluster move closer in feature space, while pixels from different clusters move apart by a fixed margin. We ignore pairs where both pixels belong to the background, because the pseudo labels cover only a subset of object instances and the remaining background may contain unlabeled objects.

Formally, let $\mathbf{z}_i$ and $\mathbf{z}_j$ denote the normalized learned feature vectors at sampled pixels i and j , and let y_i and y_j denote their pseudo instance labels. We treat the largest pseudo-label in each frame as background and assign it label 0. We define the pairwise objective as

$$\mathcal{L}(i, j) = \begin{cases} \|\mathbf{z}_i - \mathbf{z}_j\|_2^2, & y_i = y_j \neq 0, \\ \max(m - \|\mathbf{z}_i - \mathbf{z}_j\|_2, 0)^2, & y_i \neq y_j, \\ 0, & y_i = y_j = 0, \end{cases} \quad (4)$$

where the margin is set to $m = 1.0$. Let $\mathcal{P}$ denote all pairs formed from the 200 sampled pixels after excluding background-background pairs. The total training loss is

$$\mathcal{L}_{\text{total}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \mathcal{L}(i, j). \quad (5)$$

This objective encourages the encoder to capture object unity and instance separation without explicitly supervising category identity. Although video clips provide the supervision during pretraining, the encoder itself takes only a single image as input. After pretraining, it can therefore serve as an ordinary image backbone without optical flow. The encoder learns appearance cues for recovering object-level grouping from a static image.

3.3 Large-Scale Data Processing

Data sources. We use four video sources for pretraining. OpenDV-YouTube [58] contains front-view driving videos collected from more than 244 cities, with about 1,747 hours of raw video. nuPlan [59] provides 1,200 hours of driving logs from four cities, including 120 hours with eight camera views. We further add 1,700 hours from NVIDIA PhysicalAI-Autonomous-Vehicles [60] and 2,516 hours of self-collected videos. In total, the pretraining corpus contains 7,163 hours of raw video. The same video sources are used in both training cycles; Cycle 2 increases the number of usable pseudo-labeled frames rather than introducing new sources.

Optical flow estimation. We process all sources with the same video-to-flow pipeline. We decode videos into short clips and run VideoFlow [61], a multi-frame flow estimator, around each target frame. Each clip contains five frames, and VideoFlow estimates optical flow for the middle frames. We then compute a valid-flow mask using a forward-backward consistency check. Let $\mathbf{w}_f = \mathbf{w}_{t \rightarrow t+1}$ denote the forward flow and $\mathbf{w}_b = \mathbf{w}_{t+1 \rightarrow t}$ denote the backward flow. A pixel is considered valid when the forward-backward residual is smaller than a flow-magnitude-aware threshold:

$$\begin{aligned} p' &= p + \mathbf{w}_f(p), & r(p) &= \|\mathbf{w}_f(p) + \mathbf{w}_b(p')\|_2, \\ \tau(p) &= \alpha \left(\|\mathbf{w}_f(p)\|_2 + \|\mathbf{w}_b(p')\|_2 \right) + \beta, \\ M(p) &= \mathbf{1}\{r(p) < \tau(p)\}. \end{aligned} \quad (6)$$

where $M(p)$ is the valid-flow mask and (α, β) are tolerance parameters. For corpus-scale storage, we encode each flow field and its valid mask as a compact 16-bit RGB PNG: the first two channels store linearly quantized horizontal and vertical flow values after clipping to a fixed range, and the third channel stores the valid-flow mask.

Pseudo-mask construction and filtering. We construct pseudo instance masks from the stored flow fields and valid-flow masks. Starting from each unvisited valid pixel, breadth-first search traverses its 4-connected neighbors and assigns a neighboring pixel to the same cluster when the Euclidean difference between their flow vectors is no greater than $\theta_f = 1.5$. We discard clusters containing fewer than $\theta_s = 100$ pixels. The complete procedure is provided in Appendix A. Figure 3 shows representative input frames, their optical flow, and the resulting pseudo labels. We then apply several quality filters to remove unreliable pseudo labels. We retain only frames whose forward-backward valid-flow ratio exceeds 85%. Because the training loss treats the largest pseudo-label as background, we impose a left-to-right connectivity check on this label. Frames that do not satisfy this condition are discarded, reducing the risk that an erroneous flow cluster is used as the background label. Finally, we retain frames whose pseudo masks contain at least one foreground region. After processing and filtering, Cycle 1 yields 195M pseudo-labeled training frames, compared with 48M frames in the conference version.

3.4 Motion-Verified Self-Training

The cycle-1 label-generation pipeline uses motion cues in a conservative way: a pseudo instance is retained only when the flow field supports a sufficiently complete and reliable motion-induced region. In practice, many frames contain useful but incomplete motion evidence. For example, an object boundary may be partially missing, locally noisy, or connected to nearby regions, preventing BFS clustering from producing a clean closed mask. Such frames may fail to produce complete pseudo-instance masks in Cycle 1, even though their local motion cues still provide valuable information about object separation. Motion-Verified Self-Training is designed to recover this unused supervision by using the cycle-1 encoder to propose complete object masks, while motion cues are reused to verify and refine these proposals.

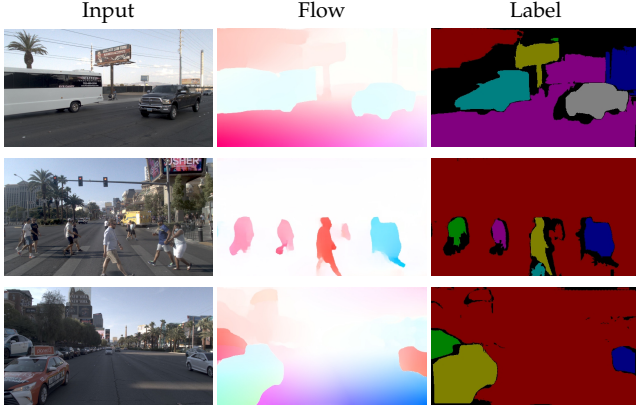


Fig. 3. Examples of Cycle-1 pseudo-label generation. We estimate optical flow from neighboring frames and cluster reliable pixels with coherent motion into category-agnostic pseudo instance labels. Colors distinguish motion clusters, not semantic categories.

Proposal generation. Figure 4 summarizes Cycle 2. We use the cycle-1 Swin-H encoder pretrained on 195M motion-derived pseudo-labeled frames. We freeze the encoder and train a MaskFormer head [62] on top of its multiscale features. Together, they form a MaskFormer proposal model that predicts candidate masks from image appearance. We apply this model to the same video sources used in Sec. 3.3, including frames for which conservative motion clustering does not yield complete pseudo-instance masks. Thus, Motion-Verified Self-Training increases label coverage from the original videos rather than introducing additional data sources.

Motion-guided refinement and verification. Inspired by the data-engine design of SAM 2 [10], we do not treat self-training as simply accepting the model’s own predictions. Instead, we introduce motion as an additional supervisory signal so that model-generated masks are refined and verified before they become new labels. We first remove low-confidence pixels, components with excessive low-confidence area, and tiny components. Let P denote a retained mask proposal and $\mathbf{z}(p)$ the dense learned feature at pixel p . We obtain a reliable interior seed by adaptive erosion,

$$S_P = \text{Erode}(P, r_P), \quad \mu_P = \frac{1}{|S_P|} \sum_{p \in S_P} \frac{\mathbf{z}(p)}{\|\mathbf{z}(p)\|_2}, \quad (7)$$

where r_P is estimated from the distance transform of P . Starting from $S_P^{(0)} = S_P$, we grow the seed within a local proposal neighborhood according to feature similarity, while preventing it from crossing strong flow-magnitude edges. With $\mathbf{w}$ denoting optical flow, let $a_w(p) = \|\mathbf{w}(p)\|_2$ be its magnitude. We define the flow-magnitude-edge indicator

$$E_w(p) = \mathbf{1} [\|\nabla a_w(p)\|_2 \geq \tau_{\text{edge}}]. \quad (8)$$

For a candidate pixel, we use the cosine similarity

$$s_P(p) = \frac{\mathbf{z}(p)^\top \mu_P}{\|\mathbf{z}(p)\|_2 \|\mu_P\|_2}. \quad (9)$$

A pixel is admissible if it is feature-compatible with the proposal and does not lie on a flow-magnitude edge:

$$\mathcal{A}_P = \{p : s_P(p) \geq \tau_{\text{feat}}, E_w(p) = 0\}. \quad (10)$$

At iteration k , the refined region is updated by

$$S_P^{(k+1)} = S_P^{(k)} \cup \left(\mathcal{N}(S_P^{(k)}) \cap \mathcal{A}_P \right), \quad (11)$$

where $\mathcal{N}(\cdot)$ denotes the 8-connected neighborhood and τ_{feat} is a fixed feature-similarity threshold. Thus, model features complete object regions from appearance cues, while flow discontinuities constrain the completion.

After refinement, we verify the resulting proposal $\tilde{P}$ using flow support. A reliable mask should have boundaries supported by nearby flow-magnitude edges and should contain limited magnitude edges in its interior. Let $\partial\tilde{P}$ be the one-pixel boundary of the mask and $\mathcal{N}_{r_{\text{align}}}(E_w)$ denote the flow-magnitude-edge map dilated by an alignment radius r_{align} . We compute

$$\begin{aligned} \rho_{\text{bdry}}(\tilde{P}) &= \frac{|\partial\tilde{P} \cap \mathcal{N}_{r_{\text{align}}}(E_w)|}{|\partial\tilde{P}|}, \\ \rho_{\text{int}}(\tilde{P}) &= \frac{|\text{Erode}(\tilde{P}, 1) \cap E_w|}{|\tilde{P}|}. \end{aligned} \quad (12)$$

The proposal is accepted only when its boundary has sufficient flow-magnitude-edge support and its interior contains limited flow-magnitude conflict. Masks with weak boundary support or high interior conflict are rejected. Finally, we merge back cycle-1 motion-derived foreground labels that are not covered by the refined model proposals. This preserves reliable cycle-1 supervision while allowing model proposals to expand label coverage. The label-generation hyperparameters are summarized in Appendix B.

After motion-guided refinement and verification, Motion-Verified Self-Training expands the training set from 195M cycle-1 pseudo-labeled frames to 421M cycle-2 pseudo-labeled frames on the same video sources. We then train the cycle-2 Swin-H encoder on this expanded set. Figure 7 shows representative label expansion, with downstream effects, coverage, and agreement analyzed in Sec. 4.7.

4 EXPERIMENTS

4.1 Implementation Details

Cycle-1 representation pretraining. We train a Swin-H encoder from scratch on 195M motion-derived pseudo-labeled frames for 50 epochs. Training uses 224×224 crops, a global batch size of 16,384, and the Muon optimizer [63] with linear warmup followed by cosine decay. The encoder is trained with the pixel-level pairwise objective described in Sec. 3.2.

Cycle-2 label generation and training. Starting from the Cycle-1 checkpoint, we freeze the Swin-H encoder and train a class-agnostic MaskFormer head [62] to generate candidate object masks. After motion-guided refinement and verification, we initialize a second Swin-H encoder from the Cycle-1 checkpoint and fine-tune it on the resulting 421M pseudo-labeled frames using the same representation objective and augmentations.

Compact-model distillation. Following the teacher-student knowledge-distillation paradigm [64], we use the Cycle-2 Swin-H encoder as a frozen teacher and distill its dense representation into randomly initialized Swin-T/S/B/L students. The distillation objective minimizes the

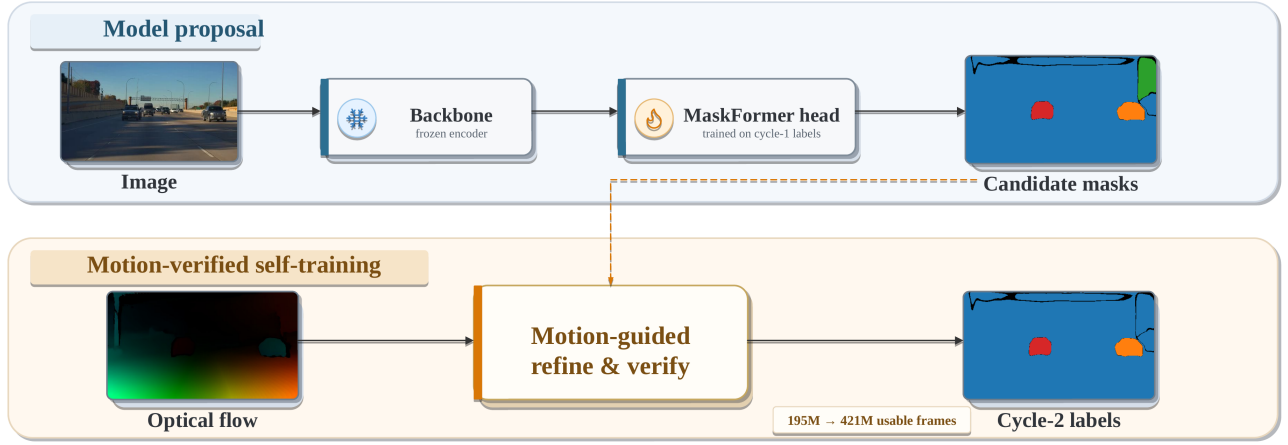


Fig. 4. Motion-Verified Self-Training. We freeze the cycle-1 Swin-H encoder and train a MaskFormer head on cycle-1 pseudo labels. The resulting proposal model generates candidate masks, which are not accepted directly: optical flow is reused as an independent signal to refine and verify them, producing cycle-2 pseudo labels and expanding the training set from 195M to 421M pseudo-labeled frames on the same video sources.

pixel-wise cosine distance between normalized teacher and student features. Only the student backbone is retained for downstream transfer.

Downstream transfer. Unless otherwise specified, *Ours* denotes the Cycle-2 model family: Swin-H is trained on 421M motion-verified pseudo-labeled frames, while Swin-T/S/B/L are distilled from the Cycle-2 Swin-H encoder. *ImageNet-22K* denotes supervised pretraining on the ImageNet hierarchy [65]. For all downstream tasks, only the pre-trained backbone is transferred; task-specific components are initialized and trained separately using the corresponding official codebase. Results marked as the *conference version* use checkpoints trained on 48M pseudo-labeled frames in our preliminary work [9]. Unless a caption or row explicitly attributes a value to a cited publication, results are produced with our downstream evaluation pipeline. The published-method rows in Table 2 are taken from their cited sources, and the planning baselines in Table 6 are taken from Xu et al. [66]; the remaining rows are our evaluations.

4.2 Qualitative Results

As already suggested by the feature-map comparison in Fig. 1, our motion-derived representation organizes visual evidence at the object-instance level rather than at the category or texture level. To inspect this behavior more locally, we compute reference-point similarity maps by measuring the cosine similarity between each colored reference feature and all spatial features in the same image.

Figure 5 shows that the responses stay concentrated on coherent object regions in these scenes, instead of spreading broadly across pixels with similar appearance. This mirrors the qualitative pattern in Fig. 1: generic foundation models such as DINOv3, CLIP, and MAE tend to emphasize category-level or texture-level structure, whereas our representation better preserves the unity of each object instance.

4.3 Monocular Depth Estimation

We first evaluate whether the learned representation transfers to low-level geometric prediction. We adopt

DCDepth [67] as the depth decoder and evaluate on the KITTI Eigen split [68], [69]. We compare our models with supervised ImageNet-22K pretraining, Semantic-SAM [70], SimMIM [49], DINOv2 [6], and DINOv3 [51]. We use SILog as the primary metric and report the remaining standard depth metrics in Table 1.

As shown in Table 1, our Swin-L obtains the best overall results, outperforming DINOv3 ViT-L in SILog (5.736 vs. 5.862), Abs Rel (0.042 vs. 0.044), Sq Rel (0.100 vs. 0.107), and RMSE (1.715 vs. 1.787). Compared with the conference Swin-L model, our current Swin-L reduces SILog from 6.558 to 5.736, an improvement of 12.5%.

Our Swin-H encoder obtains a SILog of 5.843 and an RMSE of 1.747. Its representation also transfers effectively to compact backbones through distillation: our Swin-T, Swin-S, and Swin-B obtain SILog scores of 6.714, 6.118, and 5.962, respectively. The Swin-T and Swin-B models outperform the DINOv3 ViT-S and ViT-B baselines at 7.202 and 6.477.

We further evaluate on the KITTI official online benchmark. Table 2 includes the conference Swin-L result as a reference and compares our current Swin-L model with representative published methods under the official evaluation protocol. The current model achieves a SILog of 8.07 and improves over the conference model across all four leaderboard metrics.

To inspect the geometric coherence of the predicted depth, we back-project each depth map into an RGB-colored point cloud and render it from a virtual camera translated laterally by 1.5 m (Fig. 6). This rendering makes local depth inconsistencies visible as distorted, fragmented, or spurious surfaces. Compared with DINOv2 ViT-L, our Swin-L better preserves coherent surfaces while reducing foreground-background leakage. This behavior is consistent with our motion-derived supervision, which encourages feature coherence within an object and separation across object boundaries.

4.4 3D Object Detection

We evaluate our learned visual representations on the nuScenes validation set [76] for 3D object detection, us-



Fig. 5. Reference-point similarity maps of our Swin-H encoder. Each panel places four colored reference points on the input image and visualizes, around the image, the cosine-similarity map induced by each point. High responses remain localized around coherent object regions, indicating that the learned features encode instance-level affinity rather than only category-level appearance.

TABLE 1
Monocular depth estimation on the KITTI Eigen split using DCDepth [67]. Best results are bolded.

Pretraining	Backbone	SILog ↓	Abs Rel ↓	Sq Rel ↓	RMSE ↓	RMSE log ↓	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
DINOv2 [6]	ViT-S	7.119	0.052	0.158	2.153	0.079	0.974	0.997	0.999
DINOv3 [51]	ViT-S	7.202	0.052	0.157	2.161	0.080	0.974	0.997	0.999
DINOv2 [6]	ViT-B	6.734	0.050	0.144	2.081	0.075	0.976	0.998	0.999
DINOv3 [51]	ViT-B	6.477	0.047	0.129	1.958	0.071	0.981	0.998	0.999
DINOv2 [6]	ViT-L	6.357	0.048	0.130	1.971	0.072	0.980	0.998	0.999
DINOv3 [51]	ViT-L	5.862	0.044	0.107	1.787	0.065	0.986	0.998	0.999
ImageNet-22K	Swin-T	7.455	0.055	0.165	2.182	0.082	0.969	0.996	0.999
Semantic-SAM [70]	Swin-T	7.346	0.055	0.165	2.169	0.082	0.971	0.996	0.999
ImageNet-22K	Swin-L	6.891	0.051	0.145	2.044	0.076	0.977	0.997	0.999
Semantic-SAM [70]	Swin-L	6.713	0.049	0.137	2.007	0.074	0.979	0.998	1.000
SimMIM [49]	Swin-L	6.542	0.048	0.130	1.941	0.073	0.979	0.998	0.999
Ours	Swin-T	6.714	0.049	0.134	1.958	0.075	0.978	0.998	0.999
Ours	Swin-S	6.118	0.044	0.110	1.803	0.067	0.985	0.998	1.000
Ours	Swin-B	5.962	0.043	0.105	1.766	0.065	0.986	0.998	0.999
Ours (conference)	Swin-L	6.558	0.047	0.129	1.929	0.072	0.981	0.997	0.999
Ours	Swin-L	5.736	0.042	0.100	1.715	0.063	0.988	0.999	0.999
Ours	Swin-H	5.843	0.043	0.104	1.747	0.064	0.987	0.999	0.999

TABLE 2
Monocular depth estimation on the KITTI official online benchmark. Metrics follow the leaderboard convention. Both our conference and current models use Swin-L backbones.

Method	Pretraining	Abs Rel ↓	Sq Rel ↓	iRMSE ↓	SILog ↓
NeW CRFs [71]	ImageNet sup.	8.37	1.83	11.03	10.39
VA-DepthNet [72]	ImageNet sup.	7.96	1.66	10.44	9.63
IEBins [73]	MIM [74]	7.82	1.60	10.68	9.84
NDDepth [75]	MIM [74]	7.75	1.59	10.62	9.62
DCDepth [67]	Semantic-SAM [70]	7.83	1.54	10.12	9.60
DCDepth	Ours (conference)	7.76	1.55	10.37	9.54
DCDepth	Ours	6.44	1.07	8.66	8.07

ing BEVFormer V2 [77] as the detection framework. All experiments use two input frames and a 24-epoch downstream schedule. We compare our method with supervised ImageNet-22K pretraining and the self-supervised

SimMIM [49]. All nuScenes detection metrics in Tables 3 and 4 are the official evaluator outputs multiplied by 100 for display; for example, NDS 55.89 denotes 0.5589, and mATE 62.79 denotes 0.6279 m in the evaluator’s native scale.

As shown in Table 3, our Swin-L achieves 55.89 NDS and 47.59 mAP, improving upon the conference Swin-L model at 55.80 NDS and 47.29 mAP. It also outperforms the ImageNet-22K and SimMIM baselines with the same Swin-L backbone, which obtain 54.59/45.91 and 54.98/46.52 NDS/mAP, respectively.

Our Swin-H obtains the strongest overall detection results, reaching 57.48 NDS and 49.53 mAP while yielding the lowest mATE, mASE, mAOE, and mAVE. Its representation also transfers effectively to compact backbones through distillation. Our Swin-S reaches 54.35 NDS and 45.16 mAP,



Fig. 6. Qualitative comparison of depth geometry under a 1.5 m lateral novel-view rendering. Our Swin-L produces more coherent surfaces and fewer depth artifacts than DINOv2 ViT-L.

TABLE 3
Quantitative evaluation of BEVFormer V2 [77] on the nuScenes validation set using different pretraining methods.

Backbone	Pretraining	NDS $\uparrow$	mAP $\uparrow$	mATE $\downarrow$	mASE $\downarrow$	mAOE $\downarrow$	mAVE $\downarrow$	mAAE $\downarrow$
Swin-T	ImageNet-22K	51.69	42.12	67.69	28.07	38.39	40.89	18.68
Swin-S	ImageNet-22K	53.62	45.22	64.94	27.75	36.97	40.07	20.18
Swin-B	ImageNet-22K	53.98	45.48	65.94	28.10	35.82	38.75	19.01
Swin-B	SimMIM [49]	54.03	45.18	63.81	27.53	38.02	37.04	19.22
Swin-L	ImageNet-22K	54.59	45.91	65.39	27.44	34.31	37.64	18.87
Swin-L	SimMIM [49]	54.98	46.52	64.80	28.06	33.72	35.87	20.36
Swin-T	Ours	51.56	42.49	65.21	27.97	45.17	39.47	19.01
Swin-S	Ours	54.35	45.16	64.56	27.34	35.03	35.66	19.68
Swin-B	Ours	55.69	47.67	63.12	27.70	32.96	37.96	19.75
Swin-L	Ours (conference)	55.80	47.29	62.83	27.16	33.20	36.50	18.77
Swin-L	Ours	55.89	47.59	62.79	27.49	33.30	35.60	19.84
Swin-H	Ours	57.48	49.53	61.60	27.13	31.51	33.46	19.18

while our Swin-B reaches 55.69 NDS and 47.67 mAP, outperforming both ImageNet-22K and SimMIM pretraining with the same Swin-B backbone.

To compare against ViT-based representations at a lower input resolution, we additionally evaluate all methods at 704×256 for 24 epochs. As shown in Table 4, our Swin-L reaches 52.88 NDS and 43.26 mAP, surpassing DINOv2 ViT-L by 0.97 NDS and 1.21 mAP. Our Swin-S, Swin-B, and Swin-L models also outperform their ImageNet-22K-pretrained counterparts in both NDS and mAP under this setting.

4.5 3D Occupancy Prediction

We evaluate 3D occupancy prediction on the nuScenes validation set using SparseOcc [78]. All experiments use eight input frames and a 24-epoch downstream schedule. Table 5 combines representative conference-version baselines with our extended models.

Our Swin-L obtains 40.04 RayIoU, improving upon the conference Swin-L model at 38.70 and outperforming the ImageNet-22K and SimMIM baselines with the same backbone at 37.60 and 38.60, respectively. It also surpasses DINOv2 ViT-L at 39.00, while remaining within 0.98 points of DINOv3 ViT-L, which achieves the best overall result of 41.02.

Our Swin-H obtains 39.98 RayIoU. Its representation transfers effectively through distillation: our Swin-S reaches 38.45 RayIoU, exceeding ImageNet-22K pretraining with the same backbone and DINOv3 ViT-S by 1.65 and 1.72 points, respectively. Our Swin-B achieves 39.41 RayIoU, outperforming ImageNet-22K, SimMIM, and DINOv3 ViT-B pretraining by 1.81, 1.41, and 0.62 points, respectively.

4.6 End-to-End Planning

Beyond perception, we evaluate whether our object-centric representation transfers to end-to-end planning. We integrate our Swin-L encoder into DriveSuprim [79] and

TABLE 4
Quantitative evaluation of BEVFormer V2 [77] on the nuScenes validation set at an input resolution of 704×256 .

Method	Backbone	NDS $\uparrow$	mAP $\uparrow$	mATE $\downarrow$	mASE $\downarrow$	mAOE $\downarrow$	mAVE $\downarrow$	mAAE $\downarrow$
DINOv2 [6]	ViT-S	46.24	34.88	71.65	28.45	49.97	42.70	18.84
DINOv2 [6]	ViT-B	49.08	38.36	69.74	28.21	41.81	42.40	18.84
DINOv2 [6]	ViT-L	51.91	42.05	65.04	27.35	36.45	43.79	18.51
ImageNet-22K	Swin-T	47.42	36.34	70.90	28.40	48.36	40.47	19.40
Ours	Swin-T	46.52	36.45	72.35	28.81	54.93	41.80	19.18
ImageNet-22K	Swin-S	48.78	38.00	70.65	28.35	41.20	42.95	19.01
Ours	Swin-S	50.76	40.54	68.06	28.15	40.82	39.47	18.64
ImageNet-22K	Swin-B	50.42	40.71	68.56	27.80	40.60	42.81	19.52
Ours	Swin-B	51.94	42.23	67.15	27.60	38.53	38.56	19.94
ImageNet-22K	Swin-L	50.48	40.09	68.01	27.90	40.69	40.67	18.38
Ours	Swin-L	52.88	43.26	66.79	27.86	35.76	38.53	18.61

TABLE 5
Quantitative evaluation of SparseOcc [78] on the nuScenes validation set using different pretraining methods.

Backbone	Pretraining	RayIoU $\uparrow$	RayIoU _{1m} $\uparrow$	RayIoU _{2m} $\uparrow$	RayIoU _{4m} $\uparrow$
ViT-S	DINOv2 [6]	35.90	29.50	36.80	41.40
ViT-S	DINOv3 [51]	36.73	30.40	37.56	42.24
ViT-B	DINOv2 [6]	37.10	31.00	37.90	42.40
ViT-B	DINOv3 [51]	38.79	32.65	39.71	44.01
ViT-L	DINOv2 [6]	39.00	32.80	39.90	44.30
ViT-L	DINOv3 [51]	41.02	34.93	42.12	46.00
Swin-T	ImageNet-22K	35.50	29.40	36.30	40.90
Swin-S	ImageNet-22K	36.80	30.40	37.60	42.30
Swin-B	ImageNet-22K	37.60	31.30	38.40	43.10
Swin-B	SimMIM [49]	38.00	31.70	38.70	43.40
Swin-L	ImageNet-22K	37.60	31.40	38.40	43.00
Swin-L	SimMIM [49]	38.60	32.60	39.40	43.70
Swin-T	Ours	36.72	30.50	37.52	42.13
Swin-S	Ours	38.45	32.40	39.35	43.59
Swin-B	Ours	39.41	33.13	40.35	44.74
Swin-L	Ours (conference)	38.70	32.60	39.50	43.80
Swin-L	Ours	40.04	33.99	41.09	45.02
Swin-H	Ours	39.98	33.71	40.98	45.24

evaluate under NAVSIMv2 [80], which extends the original NAVSIM benchmark [81]. Following the setup of Xu et al. [66], all compared variants use the same planning decoder and training configuration and differ only in the visual backbone. We report the extended predictive driver model score (EPDMS) using the corrected NAVSIM evaluator.¹

As shown in Table 6, our Swin-L obtains 88.9 EPDMS, outperforming DINOv2-L by 1.7 points and remaining within 0.1 points of DINOv3-L. DA-ViT-L achieves the highest aggregate score of 90.5, leaving a gap of 1.6 points to our model.

At the metric level, our model obtains the best no-at-fault collision (NC) score of 98.6 and the best time-to-collision (TTC) score of 98.1 among the compared backbones. It also matches the best history-comfort score of 98.3 and improves extended comfort over DINOv2-L and DINOv3-L. In contrast, DA-ViT-L performs better on drivable-area compliance, ego progress, lane keeping, and extended comfort, which helps explain its higher aggregate EPDMS. Overall, our model transfers competitively to end-to-end planning, with its strongest relative performance concentrated on collision-related metrics.

4.7 Effect of Motion-Verified Self-Training

We evaluate the effect of Motion-Verified Self-Training by comparing the cycle-1 and cycle-2 Swin-H encoders under

the same downstream settings. As summarized in Table 7, Cycle 2 improves all primary metrics: SILog decreases from 5.90 to 5.84 on KITTI depth estimation, NDS and mAP increase from 56.80 to 57.48 and from 48.96 to 49.53 on nuScenes 3D detection, and RayIoU increases from 39.71 to 39.98 on nuScenes occupancy prediction. With the backbone architecture and downstream protocols held fixed, these comparisons measure the net effect of the second training cycle, including its expanded motion-verified supervision.

Control for additional pretraining. The Swin-H comparison above measures the complete Cycle-2 pipeline, which changes the pseudo-label set and adds further optimization. To test whether the gain can be explained simply by training longer on the original Cycle-1 labels, we conduct a smaller controlled experiment with Swin-T. We compare the 50-epoch Cycle-1 model with a longer 75-epoch Cycle-1 run and with a Cycle-2 branch initialized from the 50-epoch Cycle-1 checkpoint and trained for 10 epochs on the motion-verified labels. The backbone, optimizer, representation objective, augmentations, batch size, and input resolution are held fixed across the two continuation branches. All three encoders are transferred using exactly the same DCDepth architecture, training schedule, and evaluation protocol. The additional Cycle-1 continuation processes approximately 4.88B samples ($25 \times 195\text{M}$), whereas the Cycle-2 branch processes approximately 4.21B samples ($10 \times 421\text{M}$), giving Cycle 2 a slightly smaller additional pretraining budget.

As shown in Table 8, extending Cycle-1 pretraining from 50 to 75 epochs does not yield a consistent improvement: SILog changes from 6.9709 to 6.9885, while Abs Rel changes

1. The evaluator correction is documented in <https://github.com/autonomousvision/navsim/issues/151>.

TABLE 6

Planning results on NAVSIMv2 using the corrected NAVSIM evaluator. Baseline results are from Xu et al. [66]; the Ours row is our evaluation. All variants use the same DriveSuprim planning decoder and training configuration and differ only in the visual backbone. NC: no-at-fault collision; DAC: drivable-area compliance; DDC: driving-direction compliance; TL: traffic-light compliance; EP: ego progress; TTC: time-to-collision; LK: lane keeping; HC: history comfort; EC: extended comfort. Best results are bolded.

Backbone	NC↑	DAC↑	DDC↑	TL↑	EP↑	TTC↑	LK↑	HC↑	EC↑	EPDMS↑
DA-ViT-L [82]	98.4	98.6	99.6	99.8	90.5	97.8	97.0	98.3	78.6	90.5
DINOv2-L [6]	97.9	97.0	99.3	99.7	87.7	97.0	95.5	98.2	77.1	87.2
DINOv3-L [51]	98.1	97.8	99.5	99.7	89.5	97.4	96.2	98.3	77.3	89.0
Ours, Swin-L	98.6	97.1	99.5	99.7	89.4	98.1	96.2	98.3	77.6	88.9

TABLE 7

Effect of Motion-Verified Self-Training under a fixed Swin-H backbone. The pretraining-set sizes count pseudo-labeled frames. Downstream architectures, training schedules, and evaluation protocols are identical between the two cycles.

Training cycle	Pretraining supervision	KITTI SLog ↓	nuScenes NDS ↑	nuScenes mAP ↑	nuScenes RayIoU ↑
Cycle 1	Motion-derived (195M)	5.90	56.80	48.96	39.71
Cycle 2	Motion-verified (421M)	5.84	57.48	49.53	39.98

TABLE 8

Swin-T control for additional pretraining on KITTI depth estimation with DCDepth. Additional sample exposures are measured relative to the 50-epoch Cycle-1 reference. The Cycle-1 continuation and Cycle-2 branch use comparable additional pretraining budgets, while all downstream settings are identical. Best results are bolded.

Variant	Pretraining schedule	Additional exposures	Abs Rel ↓	Sq Rel ↓	RMSE ↓	RMSE log ↓	δ_1 ↑	SLog ↓
Cycle 1	50 epochs on Cycle-1 labels	—	0.0509	0.1409	2.0084	0.0769	0.976	6.9709
Cycle 1, longer	75 epochs on Cycle-1 labels	4.88B	0.0512	0.1446	2.0052	0.0773	0.973	6.9885
Cycle 2	50 Cycle 1 + 10 Cycle 2 epochs	4.21B	0.0498	0.1385	1.9766	0.0756	0.978	6.8761

from 0.0509 to 0.0512. In contrast, 10 epochs of Cycle-2 pretraining improve every non-saturated depth metric, reaching 6.8761 SLog, 0.0498 Abs Rel, and 1.9766 RMSE. Thus, the Cycle-2 improvement is not attributable merely to additional optimization on the Cycle-1 labels.

Pseudo-label coverage and agreement. Figure 7 qualitatively illustrates the label expansion produced by Motion-Verified Self-Training. In these examples, conservative flow clustering labels only a subset of the visible instances, whereas Cycle 2 recovers additional object regions by combining model proposals with motion evidence. To examine whether this expanded coverage comes from simply accepting noisier proposals, we compare the Cycle-1 and Cycle-2 pseudo labels with masks generated independently by SAM 2 [10]. SAM 2 is used only as a reference for this analysis and does not provide ground truth or training supervision. We do not expect exact instance-level agreement, as our motion-derived pseudo labels and SAM 2 automatic masks may partition the same foreground regions at different granularities. We sample 200 frames labeled by both cycles and 200 frames labeled only in Cycle 2, using one frame from each of 400 distinct videos.

As shown in Table 9, Cycle 2 increases the average number of pseudo instances from 1.6 to 2.4 and reference-foreground coverage from 13.8% to 16.6% on the paired frames. The fraction of frames containing at least one matched instance also rises from 45.0% to 54.0%, while IoU changes from 83.8% to 81.6%. On the Cycle-2-only frames, IoU reaches 81.2%, and 47.5% of frames contain at least one match. Together with the qualitative examples, these results indicate that Cycle 2 expands pseudo-label coverage while retaining high overlap with the independent reference.

4.8 Complementarity with Appearance-Based Pretraining

To examine whether motion-derived and appearance-based representations provide complementary cues, we fuse their backbone features before the DCDepth decoder. Specifically, features from the two encoders are concatenated along the channel dimension. As shown in Table 10, combining our Swin-T representation with DINOv2 ViT-S improves SLog from 6.714 to 6.622. More importantly, under the same two-encoder architecture and fusion design, Ours+DINO outperforms ImageNet+DINO, reducing SLog from 7.071 to 6.622 and RMSE from 2.117 to 1.948. This controlled comparison indicates that our motion-derived representation provides cues complementary to the appearance-based representation learned by DINOv2.

5 DISCUSSION

5.1 What Does a Visual Encoder Learn from Motion Supervision?

Motion serves as the source of supervision in our framework rather than the representation target itself. The pseudo-instance labels derived from optical flow answer a grouping question: which pixels are likely to belong to the same physical entity? They do not specify the semantic category of that entity. Accordingly, the resulting single-image encoder is not trained to represent motion direction or velocity. Instead, it learns to recover object-level organization from the appearance and context available in a static image.

This distinction is reflected in our learning objective. The pairwise loss encourages pixels within the same pseudo-instance to have similar representations and separates pixels assigned to different instances, but it establishes no explicit

TABLE 9

Pseudo-label coverage and agreement with independently generated SAM 2 reference masks. Ref. coverage measures the reference foreground covered by pseudo labels.

Subset	Labels	Inst./frame	Ref. coverage (%)	IoU (%)	Matched frames (%)
Paired frames	Cycle 1	1.6	13.8	83.8	45.0
Paired frames	Cycle 2	2.4	16.6	81.6	54.0
Cycle-2-only frames	Cycle 2	2.1	14.6	81.2	47.5

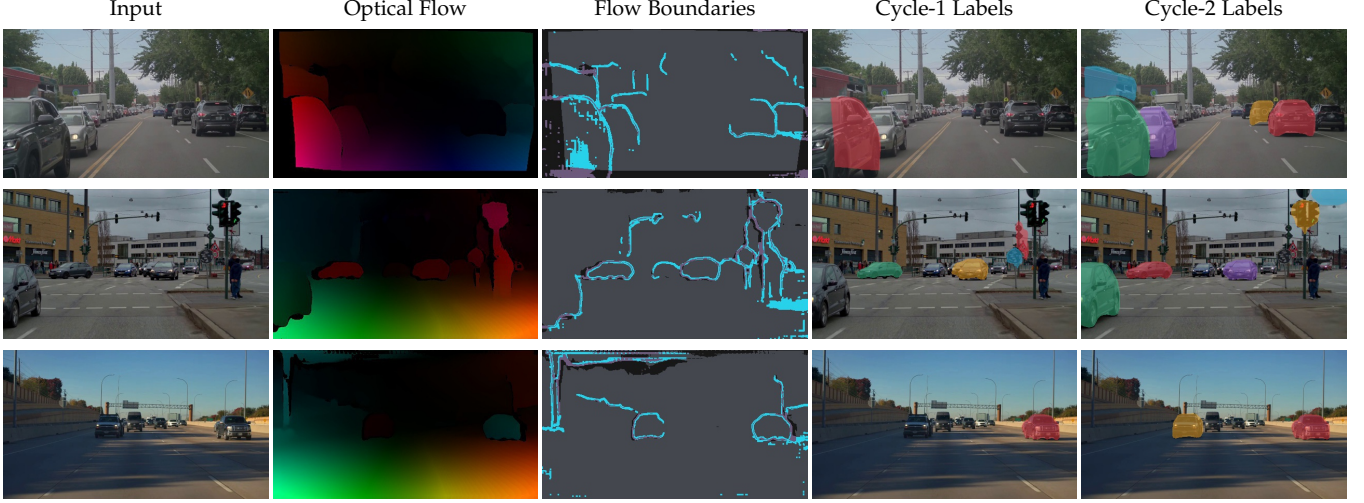


Fig. 7. Qualitative comparison of Cycle-1 and Cycle-2 pseudo labels. Each row shows an input frame, its optical flow, flow boundaries used as motion evidence, and the pseudo labels generated in the two cycles. Conservative flow clustering in Cycle 1 captures only instances supported by sufficiently complete motion regions, whereas Cycle 2 combines model proposals with motion-based verification to recover additional instances. Colors distinguish instances within each label panel and do not denote semantic categories.

TABLE 10

Feature-level complementarity on the KITTI Eigen split using DCDepth [67]. Ours and IN use Swin-T, while DINO uses DINOv2 ViT-S. For dual-encoder variants, the two feature maps are concatenated along the channel dimension before the depth decoder.

Ours	DINO	IN	SILog ↓	Abs Rel ↓	RMSE ↓
✓			6.714	0.049	1.958
	✓		7.119	0.052	2.153
		✓	7.455	0.055	2.182
	✓	✓	7.071	0.052	2.117
✓	✓		6.622	0.048	1.948

category correspondence across images. The objective therefore promotes intra-object coherence and inter-instance separation rather than category discrimination. In the examples shown in Fig. 1, our feature responses tend to preserve the unity of individual objects. This does not imply that the representation contains no semantic information; rather, its dominant inductive bias concerns object-level grouping. In this sense, motion tells the model which pixels belong together, but not what the resulting object is.

This object-centric bias helps explain the transfer performance on dense geometric and instance-sensitive perception tasks, and may also support end-to-end planning. Monocular depth estimation, 3D object detection, and occupancy prediction can all benefit from coherent object regions and clearly represented instance boundaries. For planning, distinguishing individual agents and obstacles can provide a useful visual basis for collision-sensitive decision making, consistent with our strong collision-related metrics on NAVSIMv2. Task-specific supervision can further

supply category and structural semantics and, for planning, temporal dynamics and decision objectives that are not explicitly provided by motion pretraining. Our results therefore do not suggest that motion supervision replaces category-oriented pretraining. Rather, it provides a distinct form of supervision centered on object unity and instance separation. Combining this object-centric bias with stronger category-semantic and temporal objectives is a promising direction for constructing more complete visual representations.

5.2 Limitations and Future Directions

Our framework remains dependent on the quality of optical flow. Forward-backward consistency checking and subsequent filtering remove many unreliable estimates, but they cannot recover motion evidence that is absent from the estimated flow field. Fast motion, severe occlusion, motion blur, weak texture, non-rigid deformation, and incomplete motion boundaries can all prevent the cycle-1 clustering pipeline from producing reliable pseudo-instances. Motion-Verified Self-Training improves the utilization of incomplete motion cues by combining model proposals with flow-based verification and refinement, but it cannot recover an object boundary when neither the flow field nor the cycle-1 encoder provides sufficient evidence.

This dependence also introduces a selection bias into the pretraining data. The pseudo-labeling pipeline preferentially retains frames with reliable flow and objects whose boundaries are sufficiently evident in the estimated flow. Small objects, heavily occluded regions, and objects whose

motion is difficult to distinguish from their surroundings are less likely to receive valid pseudo labels. Because the cycle-2 encoder is initialized from cycle-1 supervision, it may inherit some of these biases even as it increases label coverage. Moreover, processing hundreds of millions of frames requires substantial computation for optical-flow estimation, mask construction, and filtering.

The learned representation also reflects the granularity of its supervision. Encouraging feature consistency over an entire pseudo-instance benefits object-level grouping, but may suppress distinctions among parts within the same object. Consequently, the representation may be less suitable for tasks that require fine-grained local correspondence or part-level discrimination. Local correspondence objectives could be combined with our object-level objective to preserve both object unity and internal structure. In addition, although our supervision is derived from videos, the resulting encoder operates on individual frames and does not explicitly maintain object identity over time, model interactions among objects, or predict future dynamics. Extending the representation with temporal association and prediction remains an important direction.

In the current framework, the pretrained optical-flow model is treated as a fixed supervision generator. A promising direction is to establish a motion-to-object-to-motion bootstrapping loop. Optical flow would first generate pseudo-instance labels for learning an object-centric encoder; the resulting encoder could then initialize or augment the flow model with object-level features and boundary priors. Such priors may improve correspondence in weakly textured or partially occluded regions and reduce erroneous flow propagation across object boundaries. The improved flow model could subsequently generate more reliable motion supervision for the next generation of object-centric encoders. This iterative process could allow motion estimation and object representation learning to improve each other, offering a scalable path toward increasingly reliable supervision and richer object-centric representations learned from raw video.

6 CONCLUSION

We presented a biologically inspired framework for learning object-centric visual representations from raw videos without human annotations on the target video corpus or camera calibration. The framework converts optical flow into category-agnostic pseudo-instance supervision and transfers the resulting object-level organization to a single-image encoder through dense pairwise metric learning. By scaling the unified processing pipeline to 7,163 hours of heterogeneous video, we obtained 195M motion-derived pseudo-labeled frames. Motion-Verified Self-Training then combined appearance-based proposals with proposal-independent motion evidence to recover additional supervision from the same videos, expanding the training set to 421M frames. We further scaled the representation to a Swin-H backbone and distilled it into compact variants.

Across monocular depth estimation, 3D object detection, 3D occupancy prediction, and end-to-end planning,

the learned representations transfer competitively with supervised and self-supervised alternatives, with particularly strong transfer on geometry- and instance-sensitive tasks. The controlled Swin-T comparison further demonstrates that Cycle-2 supervision yields better depth transfer than simply extending pretraining on the Cycle-1 labels. These results support the view that motion provides a distinct form of visual supervision: it encourages object unity and instance separation without specifying semantic categories. Motion-derived pretraining therefore complements, rather than replaces, category-oriented and temporal objectives. Combining these sources of supervision, while improving the reliability and coverage of motion cues, offers a promising direction toward more complete visual representations learned from raw video.

APPENDIX A PSEUDO-CODE FOR PIXEL CLUSTERING

For each optical-flow field produced by VideoFlow, we apply breadth-first search (BFS) to group valid pixels into motion-induced clusters. Algorithm 1 takes the optical flow, the valid-flow mask obtained from the forward-backward consistency check, and two thresholds θ_f and θ_s as input. The flow threshold θ_f determines whether two 4-connected neighboring pixels have sufficiently similar motion to belong to the same cluster, while θ_s specifies the minimum cluster area.

Algorithm 1 Pixel Clustering

Input: flow, valid, θ_f , θ_s

Output: Set of retained clusters S

```

1: visited[ $x, y$ ]  $\leftarrow$  false for all pixels
2:  $S \leftarrow \emptyset$ 
3: for  $x \leftarrow 1$  to  $H$  do
4:   for  $y \leftarrow 1$  to  $W$  do
5:     if visited[ $x, y$ ] or  $\neg$ valid[ $x, y$ ] then
6:       continue
7:     end if
8:      $Q \leftarrow$  empty queue,  $C \leftarrow \emptyset$ 
9:     visited[ $x, y$ ]  $\leftarrow$  true
10:    Enqueue( $Q, (x, y)$ )
11:    while  $Q \neq \emptyset$  do
12:       $(x, y) \leftarrow$  Dequeue( $Q$ )
13:       $C \leftarrow C \cup \{(x, y)\}$ 
14:      for  $(i, j)$  in the 4-neighbors of  $(x, y)$  do
15:        if  $(i, j)$  is in bounds, valid[ $i, j$ ], and  $\neg$ visited[ $i, j$ ] then
16:          if  $\|\text{flow}[i, j] - \text{flow}[x, y]\|_2 \leq \theta_f$  then
17:            visited[ $i, j$ ]  $\leftarrow$  true
18:            Enqueue( $Q, (i, j)$ )
19:          end if
20:        end if
21:      end for
22:    end while
23:    if  $|C| \geq \theta_s$  then
24:       $S \leftarrow S \cup \{C\}$ 
25:    end if
26:  end for
27: end for
```

APPENDIX B

CYCLE-2 LABEL GENERATION HYPERPARAMETERS

Let $Q_q(\cdot)$ denote the q -th percentile, let $d_P(p)$ denote the foreground distance-transform value of proposal P at pixel p , let $a_w(p) = \|\mathbf{w}(p)\|_2$ denote the optical-flow magnitude, and define its gradient magnitude as

$$g_w(p) = \|\nabla a_w(p)\|_2. \quad (13)$$

Table 11 summarizes the hyperparameters used for motion-guided refinement and verification. Here, Ω denotes the image domain, while τ_{bdry} and τ_{int} denote the minimum boundary-support ratio and maximum interior-conflict ratio, respectively. A refined proposal $\tilde{P}$ is accepted when $\rho_{\text{bdry}}(\tilde{P}) \geq \tau_{\text{bdry}}$ and $\rho_{\text{int}}(\tilde{P}) \leq \tau_{\text{int}}$. All spatial radii are measured at the 512×288 cycle-2 label resolution.

TABLE 11
Hyperparameters for cycle-2 label generation.

Symbol	Setting
r_P	$\min(8, \lfloor Q_{40}(\{d_P(p) : p \in P\}) \rfloor)$
τ_{feat}	0.3
τ_{edge}	$\max(Q_{75}(\{g_w(p) : p \in \Omega\}), 0.3)$
r_{align}	3 pixels
τ_{bdry}	0.4
τ_{int}	0.05

APPENDIX C

ADDITIONAL FEATURE VISUALIZATIONS

Figure 8 provides additional feature-map visualizations from the cycle-2 Swin-H encoder on driving and self-collected scenes. We reduce the dense features to three dimensions using PCA and map them to RGB. The PCA colors are defined independently for each image and indicate only within-image feature similarity.

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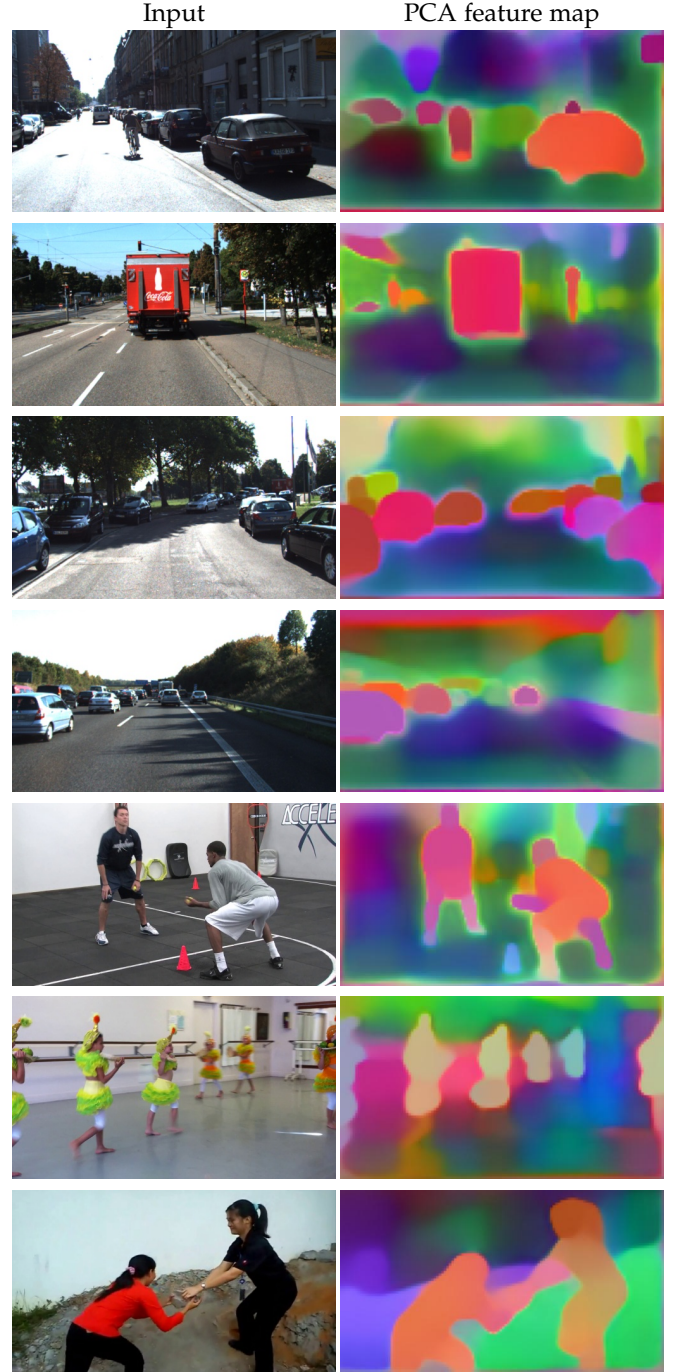


Fig. 8. Examples of the output features. Each row shows an input image and the corresponding three-component PCA visualization. PCA colors are defined independently for each image.

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